

Ultra-high Mode Purity and SMF-matched O-Band Hybrid Antiresonant Hollow Core Fiber for Intra Data Center Connection

Zihan Dong, Peng Li*, Jun Chu, Lu Dai, Hongmin Li, Yifei Xu, Guoqun Chen, Anqing Jia, Lei Zhang, Jie Luo

State Key Laboratory of Optical Fiber and Cable Manufacture Technology, Yangtze Optical Fiber and Cable Joint Stock Limited Company (YOFC), Wuhan 430074, Hubei, China.

* Email: lipeng@yofc.com

Abstract We report the first demonstration of an O-band hybrid antiresonant hollow core fiber with ~ 6000 dB/km LP_{11} mode loss and $14/240$ μm core/coating diameters. Meanwhile, this fiber delivers macrobending loss comparable to G.657.A2, rendering it well suited for high-density IDC interconnections. © 2026 The Authors

Introduction

The explosive growth of AI computing and high-performance computing has driven data center networks (DCNs) to rapidly evolve toward scale-out high-density cluster architectures, where intra-rack and inter-rack short-reach interconnections demand unprecedentedly ultra-low latency, high density, and low power consumption. The O-band has become the mainstream for DCN short-reach links thanks to the low-cost, low-power advantage of direct-modulation lasers. Conventional solid-core single mode fibers (SMFs) are inherently constrained by latency and nonlinearity limits, whereas hollow core fibers (HCFs) fundamentally break these bottlenecks—making O-band HCFs the ideal solution to meet the strict low-latency demand of AI-driven scale-out DCNs. To enable practical and reliable deployment of HCFs in this scenario, several key performance characteristics are highly desirable, including: coating diameter compatible with standard SMFs for high-density cabling, mode-field diameter

(MFD) matched to SMFs for low-loss interconnection, excellent macro-bending resistance to tolerate tight routing in compact spaces, and robust single-mode operation to ensure high-speed transmission quality.

Significant progress has been achieved in SMF-compatible HCFs based on double-nested and triple-nested antiresonant nodeless fibers (DNANF/TNANF), which deliver improved mode field diameter matching and enhanced macro-bending robustness. Nevertheless, their single mode performance remains insufficient: the LP_{11} mode losses are 400 dB/km [1] for DNANF and 95 dB/km for TNANF [2], respectively. Such levels only guarantee effectively single mode operation for fiber lengths beyond several tens of meters [1]. Per the recommendations of ITU-T G.650.1, however, excellent single mode behavior should be maintained even for lengths shorter than 10 m, imposing a much stricter demand for the LP_{11} mode loss.

In this work, we propose a hybrid antiresonant

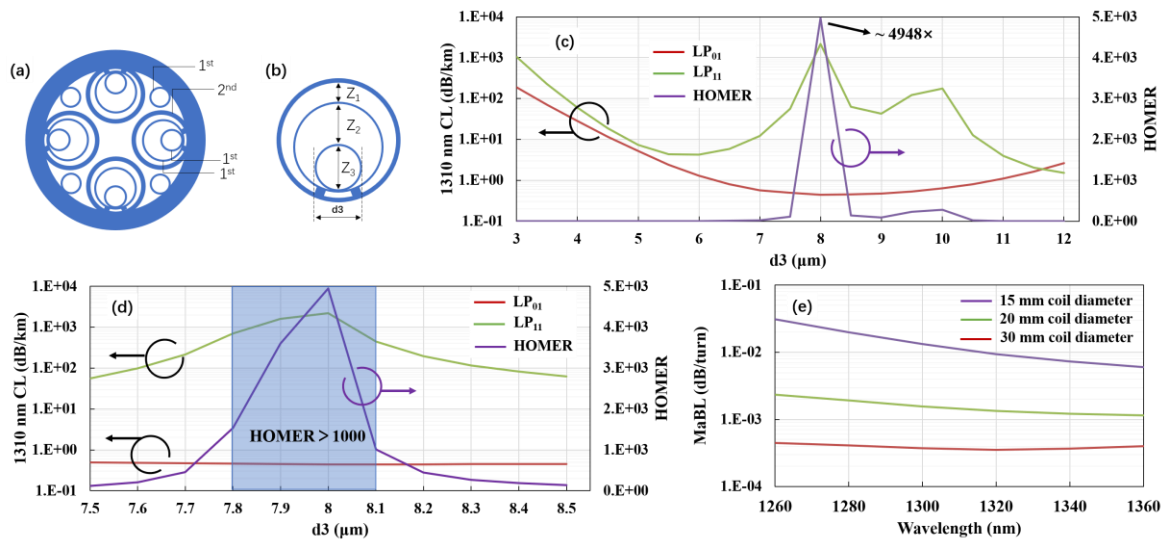


Fig. 1: Design and simulation. Schematic diagrams of (a) the HA-HCF structure and (b) an enlarged view of an antiresonant unit in the HA-HCF. The distance introduced by the supporting structure is $1 \mu\text{m}$. (c) Simulation results of LP_{01} loss, LP_{11} loss and HOMER under different values of d_3 (1310 nm). (d) Detailed simulation results in the vicinity of $d_3 = 8 \mu\text{m}$. (e) MaBL results at O-band for bending diameters of 30, 20, and 15 mm with a d_3 of $7.948 \mu\text{m}$ ($Z_2/D_{\text{core}} = 0.518$).

hollow core fiber (HA-HCF) based on a gap-tube-assisted support structure [3]. We successfully fabricated a continuous 10.037 km fiber with 240 μm coating diameter and MFD $<10\text{ }\mu\text{m}$. This fiber achieves a loss of $<1.4\text{ dB/km}$ across 1262–1340 nm, alongside an LP_{11} loss of $\sim 6000\text{ dB/km}$ and macrobending loss (MaBL) comparable to G.657.A2. Our work resolves the key fabrication and performance bottlenecks, providing a practical HCF solution for O-band short-reach DCN interconnections.

Simulation and Fabrication

The schematic diagram of the HA-HCF is shown in Fig. 1(a). In the HA-HCF, the large tubes are configured with a 2nd antiresonant thickness to achieve lower pressure sensitivity, facilitating long-distance drawing. Meanwhile, the remaining antiresonant tubes employ a 1st thickness to provide greater manufacturing tolerance, benefiting flexible structural optimization [4].

The finite element method (FEM) is employed to determine the optimal structural proportions that simultaneously achieve high single-mode performance and low MaBL. For the O-band, the thicknesses of the 1st and 2nd membranes are set to 0.4 μm and 1 μm , respectively. The core diameter (D_{core}) is fixed at 14 μm to better match with the MFD of single-mode fibers in the O-band. The outer diameter of the large tubes is set to 22 μm , leading to a gap between large tubes of approximately 3.46 μm , which effectively suppresses energy leakage through the gaps. At this point, as illustrated in Fig. 1(b), the key factors influencing the fiber performance are the distance Z_1 between the large and middle tubes, the distance Z_2 between the middle and small tubes, and the inner diameter Z_3 of the small tubes. Since Z_1 is adjacent to the core, an increase in Z_1 is prone to induce coupling loss of the LP_{01} mode in the core under fiber bending conditions. Consequently, Z_1 is fixed at 3 μm —a value that is sufficiently small relative to the core and feasible in practical fabrication. The outer diameter of the small tubes (d_3) is subsequently varied, which in turn adjusts Z_2 and Z_3 , to identify the optimal structural ratio.

Fig. 1(c) illustrates the confinement loss (CL) of the LP_{01} and LP_{11} modes under different d_3 values. When d_3 is 8 μm , the CL of the LP_{01} mode is less than 1 dB/km, and the LP_{11} mode can be effectively coupled into the Z_2 cavity, resulting in a higher-order mode extinction ratio (HOMER) of approximately 4948. Furthermore, more detailed simulation results in the vicinity of $d_3 = 8\text{ }\mu\text{m}$ are presented in Fig. 1(d). Within the d_3 range of 7.8–8.1 μm , the HA-HCF can achieve a HOMER exceeding 1000, with the corresponding Z_2 ranging from 7.1 to 7.4 μm and Z_3 ranging from 7.0 to 7.3 μm , respectively (Z_2/D_{core} ratio: 0.507–0.529).

When the Z_2/D_{core} ratio is 0.518, the simulated MaBL is presented in Fig. 1(e). In the O-band, the MaBL values at bending diameters of 30, 20, and 15 mm are less than 0.001, 0.01, and 0.1 dB/turn, respectively. Therefore, the Z_2/D_{core} ratio range of 0.507–0.529, which balances high HOMER and low MaBL, is selected as the target range for fiber fabrication.

Throughout the fiber drawing process, high furnace temperature and low tension parameters are employed to prevent fiber breaking with a 240 μm coating diameter. Owing to the robustness of the HA-HCF structure, the single-spool drawing length can reach 10.037 km, with excellent uniformity in both transverse and longitudinal directions. As illustrated in Fig. 2(b) and (d), the core diameter varies merely from 13.85 μm at the start of pull (SOP) to 13.71 μm at the end of pull (EOP). The 1st and 2nd membrane thicknesses are $0.40 \pm 0.03\text{ }\mu\text{m}$ and $1.02 \pm 0.03\text{ }\mu\text{m}$ respectively. At both the SOP and EOP, the average large tube diameter is 22.3 μm , resulting in a gap of $3.35 \pm 0.25\text{ }\mu\text{m}$. The radial distances Z_1 , Z_2 , and Z_3 within the nested elements are $3.3 \pm 0.5\text{ }\mu\text{m}$, $7.3 \pm 0.2\text{ }\mu\text{m}$, and $7.2 \pm 0.4\text{ }\mu\text{m}$, respectively (Z_2/D_{core} ratio: 0.51–0.55). The gap-assisted tubes have a diameter of $7.4 \pm 0.2\text{ }\mu\text{m}$, and the distance between gap tube and the large tube is $2.9 \pm 0.5\text{ }\mu\text{m}$.

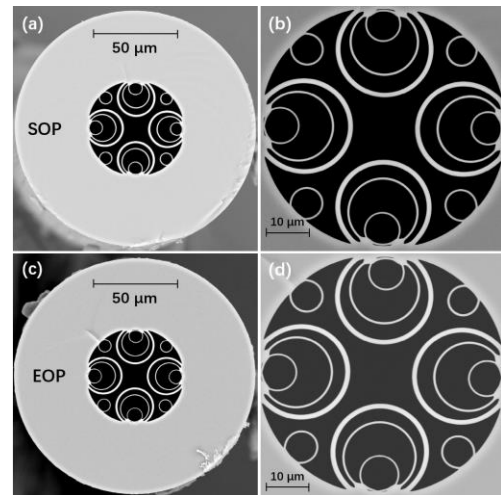


Fig. 2: SEM image. (a) and (b) show the overall image and the magnified view of the microstructure region at the SOP, respectively. (c) and (d) are the corresponding EOP.

Characterization

The loss of the HA-HCF was characterized via the cutback method, in which a supercontinuum source and an optical spectral analyzer (OSA) were utilized. The tested fiber was cut from 10.037 km to 10 m. At each length, the output end face was cleaved three times, and the transmission spectrum was averaged from the measurements obtained after each cleavage. Fig. 3(a) demonstrates that the fiber exhibits a loss of $<1.4\text{ dB/km}$ in the wavelength range of 1262–1340 nm,

with a minimum loss of 1.15 dB/km at 1310 nm. This represents the lowest loss among all reported O-band HCFs with MFD and coating diameters compatible with standard SMFs. The wavelength range of 1340–1400 nm corresponds to a region severely affected by water vapor absorption peaks. Based on the fiber structure extracted from Fig. 2(b), the CL and surface scattering loss (SSL) of the fiber were simulated, and the results were compared with the measured loss. The results indicate that the total fiber loss is primarily attributed to CL and SSL, with CL accounting for the larger proportion (approximately 55–64% in the wavelength range of 1270–1330 nm). Microbending loss is negligible due to the extremely small MFD [1,2].

Since the beam spot of the HA-HCF is approximately square, there are two distinct MFDs along the diagonal and midline directions. The MFDs at the two ends of the HA-HCF, measured via the far-field scanning method, are 9.97/9.01 μm (diagonal/midline) and 9.66/8.81 μm (diagonal/midline), respectively. The MFD results in the midline direction are compatible with those of the G.657.A2 standard (8.4–9.2 μm).

The MaBL of the HA-HCF was characterized at three bending diameters (30, 20, and 15 mm) and compared with that of the G.657.A2 fiber. It should be noted that, under a fixed MFD, the MaBL of solid core fibers is proportional to the wavelength, whereas that of HCFs is inversely proportional to the wavelength [5]. The HA-HCF was tested with a length of 8 m, wound for 10 turns at each bending diameter, and directly spliced to the 1310 nm source pigtail. To minimize random measurement errors, three measurements were performed at each bending diameter, and the average results are presented in Fig. 3(b). The MaBL of the G.657.A2 fiber at 1310 nm was also measured and shown in Fig. 3(b) as a reference. At bending diameters of 30, 20, and 15 mm, the MaBLs of the HA-HCF were 0.0013,

0.0026, and 0.03 dB/turn, respectively. Compared with the G.657.A2 standard at 1550 nm, all MaBL results of the HA-HCF demonstrate significantly superior performance.

The single-mode performance of the HA-HCF was characterized via spatially and spectrally (S^2) resolved imaging, and the LP_{11} mode loss was measured using the cutback method. At a fiber length of 5 m, the LP_{11} mode was no longer observable. Therefore, the LP_{11} mode loss was measured over the range of 1–4 m. At each length, the multipath interference (MPI) of the LP_{11} mode was measured three times and averaged, with the resulting data presented in Fig. 3(c). The curve fitted to the averaged data indicates that the HA-HCF exhibits an LP_{11} mode loss of approximately 6000 dB/km, which demonstrates its exceptionally strong higher-order mode filtering capability. Thus, in application scenarios involving distances of longer than 5 meters, the HA-HCF can deliver a pure single-mode output.

Conclusion

We have successfully achieved continuous drawing of a 10-km-long HA-HCF with a MFD of <10 μm and a coating diameter of 240 μm . This HA-HCF exhibits a loss of 1.15 dB/km at 1310 nm and <1.4 dB/km in the wavelength range of 1262–1340 nm, which is the lowest reported loss for HCFs with an MFD compatible with standard SMFs. Benefiting from the balanced design of the internal structural ratio, the HA-HCF achieves a MaBL close to that of the G.657.A2 fiber under the premise of an LP_{11} mode loss of approximately 6000 dB/km. This is the first realization of an ultrashort-distance single-mode hollow-core fiber with a small mode field and small outer diameter, which is suitable for future high-density and low-latency intra-data center interconnections.

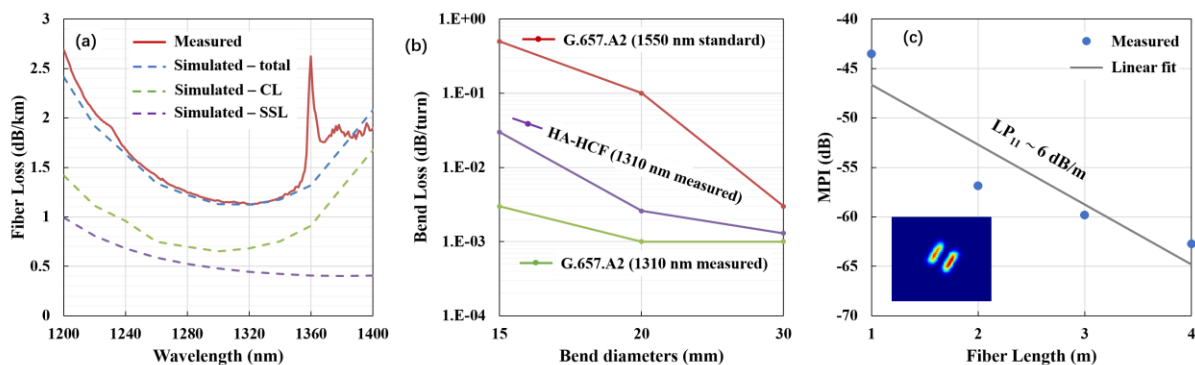


Fig. 3: (a) Cutback and simulated results of HA-HCF loss. (b) The 1310 nm MaBL of an 8 m HA-HCF at bending diameters of 30, 20, and 15 mm, compared to G.657.A2. (c) The 1310 nm MPI using the S^2 and LP_{11} loss using cutback method. The inset shows the mode pattern of LP_{11} .

Acknowledgements

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